EEZ suitable for aquaculture species

EDS 223 HW Assignment #3

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2026-11-07

This analysis aims to determine priority zones for potential aquaculture species along the West Coast based on suitable area for those species. This analysis will map priority zones for oyster and red abalone species. However, the function created can assess zones for many marine species given a species' suitable range of sea surface temperature and ocean depth.

Load packages

```
# Load libraries
library(terra)
library(tidyverse)
library(tmap)
library(sf)
```

Import data

Import shapefile of the EEZ boundaries, depth, and SST data from 2008-2012.

```
# Read in West Coast EEZ shapefile
eez <- st_read("data/wc_regions_clean.shp")
```

```
# Read in bathymetry data
depth <- rast("data/depth.tif")
```

```r
# Add SST data from 2008-2012
sst_08 <- rast("data/average_annual_sst_2008.tif")
sst_09 <- rast("data/average_annual_sst_2009.tif")
sst_10 <- rast("data/average_annual_sst_2010.tif")
sst_11 <- rast("data/average_annual_sst_2011.tif")
sst_12 <- rast("data/average_annual_sst_2012.tif")

# Combine rasters to one stack
sst_stack <- c(sst_08, sst_09, sst_10, sst_11, sst_12)
```

```r
# Check if CRSs match
if(!same.crs(eez, depth)){
  warning("Coordinate refrence systems do not match. CRS transformed to match EEZ")
  depth <- project(depth, "EPSG:4326")
}

if(!same.crs(eez, sst_stack)){
  warning("Coordinate refrence systems do not match. CRS transformed to match EEZ")
  sst_stack <- project(sst_stack, "EPSG:4326")
}
```

Process data

Sea surface temperature across 2008-2012 will be averaged to find the mean SST for that area. This average will be compared against the species' suitable SST range. Since the SST data is recorded in Kelvin, it will also be converted to Celcius.

```r
# Find mean SST across 2008-2012
sst_mean <- app(sst_stack, fun = mean, na.rm = TRUE)

# Convert SST from K to C
sst_mean <- sst_mean - 273.15
```

Next, the depth raster will be cropped to the same extent as SST data.

```r
# Crop depth raster to match extent of SST
depth <- crop(depth, ext(sst_mean))

# Compare depth resolution to SST data
cat("SST Resolution:", res(sst_mean), "\n")
```

```
SST Resolution: 0.04166185 0.04165702
```

```
cat("Depth Resolution:", res(depth), "\n")
```

```
Depth Resolution: 0.004166667 0.004166667
```

Since `sst_mean` is at a courser resolution than `depth`, `depth` will be resampled to match the SST data using the bilinear method.

```
# Resample depth to resolution of SST
depth <- resample(depth, sst_mean, method = "bilinear")

# Check resolution, extent, and CRSs match
if(!all(res(depth) == res(sst_mean))){
  warning("Resolutions do not match.")
}

if(!ext(depth) == ext(sst_mean)){
  warning("Extents do not match.")
}

if(!same.crs(depth, sst_mean)){
  warning("CRSs do not match.")
}
```

Now that both rasters match, they can be stacked together.

```
# Stack depth and mean sst
depth_sst <- c(depth, sst_mean)
```

Calculate suitable habitat and determine most suitable EEZ

With a clean raster of depth and sst data for the region, the next steps are to compare the suitable SST and depth range of species to the area. Then, priority EEZ zones for potential aquaculture will be determined based on zones with the most suitable area for the species. The function below takes the following parameters:

Parameter	Class	Description
`species`	character	The name of the species.

Parameter	Class	Description
`temp_min`	numeric	The minimum suitable temp for the species.
`temp_max`	numeric	The maximum suitable temp for the species.
`depth_min`	numeric	The minimum suitable depth for the species.
`depth_max`	numeric	The maximum suitable depth for the species.
`eez`	sf, data.frame	The shapefiles of the EEZ regions with geometries. Contains `rgn` and `rgn_key` columns.
`depth_sst_rast`	SpatRaster	A stacked raster with depth (layer 1) and SST (layer 2) data

The output is a map of the West Coast with priority EEZ regions for potential aquaculture shaded by the amount of suitable area for the given species within each zone.

```
# Define function for visualizing eez priority zones for aquaculture

priority_eez <- function(species, temp_min, temp_max,
                         depth_min, depth_max,
                         eez, depth_sst_rast
                        ) {

    # Define rcl for suitable depth & temp
    rcl_depth <- matrix(c(
        -Inf,      depth_min, 0,
        depth_min, depth_max, 1,
        depth_max, Inf,       0
    ), ncol = 3, byrow = TRUE)

    rcl_temp <- matrix(c(
        -Inf,     temp_min, 0,
        temp_min, temp_max, 1,
        temp_max, Inf,      0
    ), ncol = 3, byrow = TRUE)

    # Reclassify depth and temp areas to suitable (1) or not (0) for species
```

```r
    habitat_depth <- classify(depth_sst_rast[[1]], rcl = rcl_depth)
    habitat_temp <- classify(depth_sst_rast[[2]], rcl = rcl_temp)

    # Create raster of habitat with suitable depth and temp
    species_habitat <- habitat_depth * habitat_temp
    levels(species_habitat) <- data.frame(
        value = 0:1,
        label = c("Not Suitable", "Suitable")
    )

    # Turn suitable habitat into area km2 (instead of degrees)
    habitat_cell_size <- cellSize(species_habitat, unit = 'km')
    suitable_area = species_habitat * habitat_cell_size
    names(suitable_area) <- "suitable_area_km2"

    # Rasterize eez regions
    eez_rast <- rasterize(eez, species_habitat, field = 'rgn')

    # Calculate priority eez by most suitable area
    eez_priority = zonal(suitable_area, eez_rast, fun = 'sum', na.rm = TRUE)

    # Join geometry to eez_priority zones
    eez_priority_zones <- eez |>
        left_join(eez_priority, by = "rgn")

    # Create map for priority eez zones
    title <- paste("Priority EEZ for", species, "\nbased on suitable habitat area")
    eez_map <- tm_basemap(server = "Esri.WorldTopoMap") +
                tm_graticules(alpha=0.75) +
                tm_shape(eez_priority_zones) +
                tm_polygons(fill='suitable_area_km2',
                            fill_alpha=0.5,
                            fill.scale = tm_scale_continuous(values='brewer.blues'),
                            fill.legend = tm_legend(title="Suitable area in EEZ \n(km^2)")
                        ) +
                tm_text('rgn_key') +
                tm_title_out(title, position = tm_pos_out("center", "top", pos.v = "bottom"))
                tm_compass(position = c("left", "bottom")) +
                tm_layout(scale = 0.75)

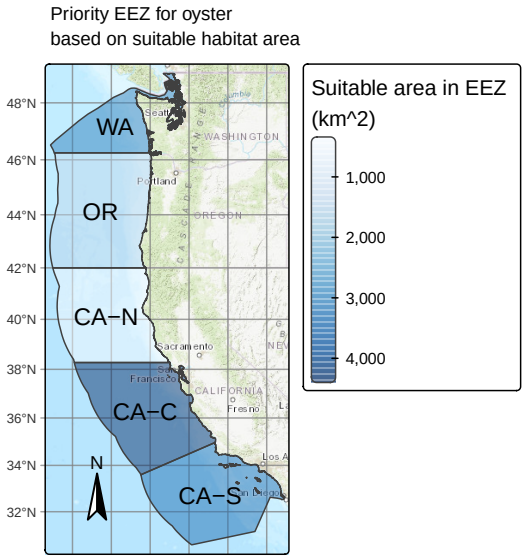
    # Return value
    return(eez_map)
```

```
}
```

Calculate priority EEZ regions for oyster aquaculture

For oysters, research has shown that optimal growing conditions for oysters are a sea surface temperature of 11-30°C and depth of 0-70 meters below sea level.

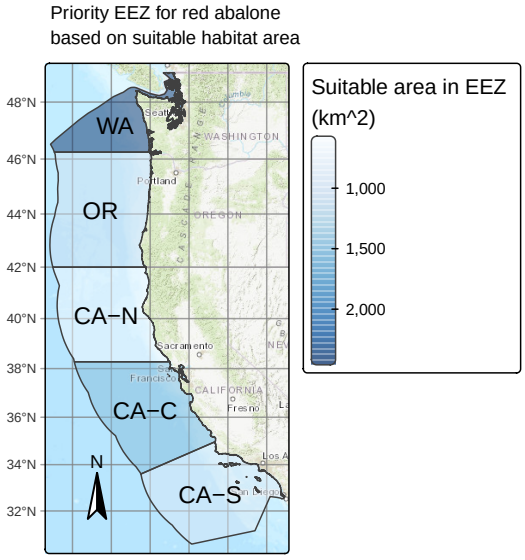
```
priority_eez(species='oyster',
                          temp_min = 11, temp_max = 30,
                          depth_min = -70, depth_max = 0,
                          depth_sst_rast = depth_sst,
                          eez = eez
                         )
```



Calculate priority EEZ regions for red abalone aquaculture

Red abalone are another potential species of interest for marine aquaculture. Their optimal conditions are a sea surface temperature between 8-18°C and a depth of 0-24 meters below sea level SeaLifeBase.

```
# Check suitable habitat for red abalone
priority_eez(species='red abalone',
                          temp_min = 8, temp_max = 18,
                          depth_min = -24, depth_max = 0,
                          depth_sst_rast = depth_sst,
                          eez = eez
                         )
```



Map Interpretation

Oysters

From the oyster map, the Central California coast appears to be the highest priority zone for potential oyster aquaculture, as this region contains more than 4,000 square kilometers of habitat with suitable sea surface temperature and depth for optimal oyster growth. The Southern Californian coast and Washington coast are the next highest priority areas, as they have close to 3,000 square kilometers of suitable area.

Red Abalone

The red abalone map suggests the Washington coast is the highest priority EEZ, with over 2,000 square kilometers of suitable area for red abalone species. The Central Californian coast ranks second with over 1,500 square km.

Map Limitations & Future Directions

It is important to note, suitable habitat was determined solely by appropriate sea surface temperature and depth. While this is an initial step towards assessing places of interest for marine aquaculture, it does not take into account other limiting factors like the species impact on the local habitat, who owns the coastline/waters with suitable habitat, and the protected status of the area (e.g. MPAs). Future directions of the project could be incorporating these other factors into determining priority locations for aquaculture at a finer scale.

Data Citations

Data	Citation	Link
EEZ shapefiles	Flanders Marine Institute. (2026). Maritime boundaries: Exclusive Economic Zones (EEZ). Marine Regions. https://www.marineregions.org/eez.php	MarineRegions
depth raster	GEBCO Bathymetric Compilation Group 2026. (2026). The GEBCO_2026 Grid: A continuous terrain model for oceans and land at 15 arc-second intervals. https://www.gebco.net/data-products/gridded-bathymetry-data#area	GEBCO
SST rasters	National Oceanic and Atmospheric Administration, Coral Reef Watch. (2018). NOAA Coral Reef Watch Daily 5km Satellite Coral Bleaching Heat Stress SST Anomaly Product (Version 3.1). https://coralreefwatch.noaa.gov/product/5km/index_5km_ssta.php	Coral Reef Watch
Sea animal ranges	Bisby, F.A., M.A. Ruggiero, K.L. Wilson, M. Cachuela-Palacio, S.W. Kimani, Y.R. Roskov, A. Soulier-Perkins and J. van Hertum 2005 Species 2000 & ITIS Catalogue of Life: 2005 Annual Checklist. CD-ROM; Species 2000: Reading, U.K. (Ref. 19)	SeaLifeBase Entry